Classification of Bt Corn from Non-Bt Corn (*Zea Mays*) Using NIR Spectral Data and Machine Learning Techniques

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***Abstract*— The growing cultivation of genetically modified (GM) crops, including *Bacillus thuringiensis* (Bt) corn, necessitates reliable and efficient methods for their detection to ensure reliable compliance and food safety. Conventional techniques like PCR and ELISA are accurate but they are also destructive, costly, and time-consuming, thus limiting their use for large-scale screening. This study explores the potential of Near-Infrared (NIR) spectroscopy, a rapid non-destructive technique, combined with machine learning for the classification of Bt and non-Bt corn kernels. The primary objective is to benchmark the performance of various machine learning algorithms which includes traditional methods such as Partial Least Squares Discriminant Analysis (PLS-DA) and Support Vector Machine (SVM), against a deep learning approach using a one-dimensional Convolutional Neural Network (1D-CNN). NIR spectral data will be collected from kernel samples, which will be processed using Savitzky-Golay smoothing. This will then be used to train and evaluate the models. Performance will be assessed based on accuracy, precision, and F1-score on a held-out test set. The results are expected to identify the most effective classification algorithm/s which will contribute to the development of a fast, cost-effective, and non-destructive tool for identifying GM corn in agricultural and industrial settings.**

***Index Terms*— Bt corn, Near-Infrared (NIR) Spectroscopy, Machine Learning, Genetically Modified Organism (GMO), Classification, *Zea Mays*, Deep Learning, Non-destructive analysis.**

I. Introduction

A. Background of the Study

Corn (*Zea Mays*) is one of the most popular crops in the world that serves as a staple food and a key source of animal feed and industrial raw materials [1]. In recent decades, genetically modified (GM) variants such as *Bacillus thuringiensis* (Bt) corn have been developed to resist pests and improve yield (Abbas, 2018). Although Bt corn offers agricultural and economic benefits, it also raises issues concerning biosafety, labeling, and trade regulations [2]. As a result, distinguishing Bt corn from non-Bt corn has become an important concern for food safety monitoring and regulatory compliance.

Previous studies have investigated several approaches to identify genetically modified crops. Conventional methods such as polymerase chain reaction (PCR) and enzyme-linked immunosorbent assay (ELISA) are accurate but both require specialized laboratory equipment, trained personnel, and time-consuming sample preparation [3], [4]. These limitations make large-scale screening impractical, especially for field or industrial use. Hence, there is growing interest in developing non-destructive, cost-effective, and efficient alternatives [4].

Near-infrared (NIR) spectroscopy has emerged as a promising analytical technique for agricultural applications because of its ability to capture subtle chemical and structural differences in biological samples [5]. When combined with modern data analysis and machine learning techniques, NIR spectral data can be used to classify and predict various crop traits such as moisture content, protein composition, and genetic characteristics. However, while there have been studies on using NIR spectroscopy in identifying genetically modified (GM) crops, only a few have focused on distinguishing Bt and non-Bt corn specifically.

In this study, an experimental approach was undertaken to identify Bt from non-Bt corn using machine learning algorithms. The study aims to evaluate which algorithm performs best in accurately differentiating between two (?) corn varieties. This paper presents the dataset collection, preprocessing technique used, the application of various machine learning models, and the evaluation of their performance in identifying Bt corn. The results are expected to contribute to the development of a rapid and non-destructive identification tool for GM crops, specifically maize.

B. Statement of the Problem

The increasing global cultivation of genetically modified (GM) crops, particularly Bt corn, has created an urgent need for reliable, efficient, and cost-effective methods for their detection. This is essential for enforcing regulatory compliance, ensuring accurate food labelling, and managing international trade. However, the current standard detection methods, which are Polymerase Chain rReaction (PCR) and Enzyme-Linked Immunosorbent Assay (ELISA), are fundamentally limited for large-scale application. These techniques are are destructive, time-consuming, expensive, and require specialized laboratory facilities and highly trained personnel. These limitations create a significant bottleneck, which makes them impractical for rapid, on-site, or high-throughput screening in agricultural and industrial settings [6].

Near-Infrared (NIR) spectroscopy has been successfully established as a viable alternative for a rapid and non-destructive technique. Numerous studies have demonstrated its effectiveness when combined with traditional machine learning models, achieving classification accuracies at or near 100%

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for various GM crops. For example, Partial Least Squares Discriminant Analysis (PLS-DA) has been used to flawlessly distinguish transgenic rice [7], and Support Vector Machine (SVM) has achieved similar success with rice and canola varieties [8], [9]. For maize specifically, both hyperspectral imaging with PLS-DA [10] and portable NIR spectroscopy with models like Extreme Learning Machine (ELM) [11] have yielded high accuracies.

More recently, however, deep learning models, specifically Convolutional Neural Networks (CNNs), have emerged as a new frontier in spectral analysis. This offers the potential for the automatic extraction of complex, hierarchical features that may outperform traditional methods. A direct comparison for GM maize kernels by Wei et al. [12] showed that a modified VGG Net (a CNN architecture) achieved 96.1% accuracy, significantly outperforming a classical backpropagation neural network. Similarly, CNNs have proven highly effective for canola classification [13]. Despite these individual successes, a comprehensive, direct benchamrking analysis that compares the gold standard chemometric models (PLS-DA, SVM) against a tailored deep learning approach (1D-CNN) for the specific task of classifying Bt corn kernels is still lacking. Without such a comparative analysis, stakeholders have no clear guidance on which computational method provides the optimal balance of accuracy, robustness, and potential for real-world deployment.

Therefore, the problem this study addresses is the need for a comparative evaluation of machine learning techniques to determine the most effective model for a practical, non-destructive Bt corn classification system. While individual methods have shown promise, the lack of a head-to-head performance analysis on this specific dataset hinders the development and adoption of an optimized and reliable tool for agricultural and industrial quality control.

C. Significance of the Study

This study is significant because it explores how machine learning can be used to identify Bt and non-Bt corn using NIR spectral data. While earlier research has shown that NIR spectroscopy can be used to classify genetically modified crops, very few studies have compared different machine learning algorithms specifically for Bt corn detection [10]–[12].

By comparing traditional algorithms such as Partial Least Squares Discriminant Analysis (PLS-DA) and Support Vector Machine (SVM) with more advanced models like Convoluted Neural Networks (CNN), this research will show which methods perform best in terms of accuracy, precision, F1 score, and interpretability [10], [12]. The comparison will also help identify the strengths and weaknesses of each approach when applied to spectral classification tasks.

The results of this study can help regulatory agencies, food testing laboratories, and agricultural companies choose the most suitable algorithms for detecting GMOs and ensuring proper labeling. In addition, the findings can contribute to the development of faster and more affordable quality control systems that can be used in agricultural and food processing industries.

Lastly, this study can serve as a reference for future research in using machine learning for agricultural data analysis by identifying which models work best and which areas still need improvement.

D. Objectives of the Study

The main objective of this study is to conduct a benchmarking analysis of various machine learning algorithms applied to NIR spectral data for classifying Bt and non-Bt corn varieties (?). The study aims to determine which models perform best in terms of classification accuracy, precision, and F1-score.

Specifically, this study seeks to:

1) Benchmark and compare the performance of traditional machine learning algorithms against deep learning approaches.
2) Develop and deploy a prototype web-based application that integrates the best-performing model, which will provide a user-friendly tool for rapid classification of unknown corn kernel samples.

E. Scope and Limitation

1) Scope: This study focuses on identifying Bt and non-Bt corn varieties using NIR spectral data. The research emphasizes benchmarking machine learning algorithms for classification.

Only NIR spectral data from maize kernels are considered, specifically transgenic Bt corn varieties and non-Bt corn. The performance of the machine learning models will be evaluated using metrics such as accuracy, precision, and F1-score.

The results of this study are expected to provide insights for regulatory agencies, food testing laboratories, and agricultural companies in selecting appropriate algorithms for efficient detection and classification of GMOs.

2) Limitations: The study has several limitations. The dataset used is specific to the maize kernel samples included in the research, and the findings may not be directly generalizable to other maize varieties or different GM crops. Only kernel samples are analyzed. Other plant parts are not considered.

The benchmarking is limited to a selected set of machine learning algorithms, and other advanced or hybrid methods are not explored. Additionally, hyperparameter tuning is constrained to feasible ranges due to computational resource limitations.

Practical factors affecting NIR spectral data, such as environmental conditions (e.g., moisture content and storage), are not fully controlled.

II. Review of Related Literature

A systematic literature review was performed to establish the current state of research in the non-destructive detection of genetically modified (GM) crops, particularly corn, using near-infrared (NIR) spectroscopy and machine learning. This review covers the foundational detection methods, the specific application of NIR spectroscopy in this domain, and the machine learning models employed for spectral classification.

A. Detection of Genetically Modified Organisms (GMOs)

The widespread cultivation of genetically modified crops, such as Bt corn, has created a regulatory and commercial need for reliable detection methods to ensure food safety, proper labeling, and compliance with trade regulations. Conventional analytical techniques for GMO detection are primarily DNA- or protein-based, including methods like polymerase chain reaction (PCR) and enzyme-linked immunosorbent assay (ELISA). While these methods are highly accurate and sensitive, they possess significant drawbacks, including being destructive, time-consuming, costly, and requiring specialized laboratory settings and trained personnel. These significant drawbacks limit their utility for rapid, large-scale, or in-field screening applications [6]. This has spurred research into alternative technologies that are non-destructive, fast, and cost-effective.

B. NIR Spectroscopy for GM Crop Detection

Near-infrared (NIR) spectroscopy has emerged as a powerful, non-destructive alternative for agricultural analysis. Operating in the 780–2500 nm range, NIR spectroscopy measures the absorption of light corresponding to overtones and combinations of molecular vibrations, particularly from hydrogen-containing functional groups (e.g., C-H, N-H, O-H) [6]. The underlying principle for its use in GM detection is that genetic modifications, even if subtle, can lead to systemic changes in a plant's chemical composition and physical structure. These alterations in metabolites like proteins, starches, oils, and moisture content create a unique "spectral fingerprint" that can be captured by NIR spectroscopy, allowing for differentiation between GM and non-GM samples [7], [10].

Successful applications of NIR and its related technique, hyperspectral imaging (HSI), have been demonstrated across various GM crops:

- **Maize:** Feng et al. [10] successfully used NIR hyperspectral imaging to discriminate transgenic maize kernels from their non-GM parent with nearly 100% accuracy. Another study by Feng et al. [11] further demonstrated the feasibility of using a portable NIR spectrometer to screen transgenic maize leaves, highlighting the potential for in-field applications.
- **Rice:** Similar success has been found with rice. Zhang et al. [7] achieved 100% classification accuracy in distinguishing transgenic rice from wild-type varieties using NIR spectroscopy combined with Partial Least Squares Discriminant Analysis (PLS-DA). Similarly, Hao et al. [8] achieved 100% accuracy in identifying transgenic rice using a Support Vector Machine (SVM) model.
- **Canola:** The technique's versatility extends to other crops like canola. Sohn et al. [13] effectively used Vis-NIR spectroscopy to discriminate between GM canola, non-GM canola, and their hybrids with nearly 99.4% accuracy.

These studies collectively establish that NIR spectroscopy is a viable and robust tool for the rapid and non-destructive identification of GM crops.

C. Machine Learning Models for Spectral Classification

Raw NIR spectral data are high-dimensional, contain significant noise, and often exhibit multicollinearity and baseline shifts. Therefore, advanced data analysis through chemometrics and machine learning is essential to extract meaningful patterns and build accurate classification models [11], [12]. A critical preliminary step is spectral preprocessing, which aims to reduce noise and correct for baseline shifts and scattering. Common preprocessing methods include Savitzky-Golay (SG) smoothing, Standard Normal Variate (SNV), Multiplicative Scatter Correction (MSC), and various derivatives [8], [9].

1) Traditional Chemometric and Machine Learning Models: Several traditional machine learning models have been successfully applied to classify NIR spectral data of GM crops.

- **Principal Component Analysis (PCA):** Often used for exploratory data analysis and dimensionality reduction, PCA can reveal initial clustering or separation between different sample groups. However, it is an unsupervised method and often insufficient for definitive classification on its own, as demonstrated by Feng et al. [10] and Hao et al. [8], where significant overlap remained between GM and non-GM clusters in the PC scores plots.
- **Partial Least Squares Discriminant Analysis (PLS-DA):** As a supervised regression method adapted for classification, PLS-DA is highly effective for handling collinear spectral data. Its successful application is demonstrated by Zhang et al. [7] and Feng et al. [10], who achieved classification accuracies of 100% and 99.00%, respectively, for rice and maize kernel identification.
- **Support Vector Machine (SVM):** SVM is a powerful supervised classifier known for its excellent performance with high-dimensional data. It has consistently proven to be one of the most accurate models in this domain. Sohn et al. [9] achieved 100% accuracy for canola classification using an SG-SVM combination. Likewise, Hao et al. [8] found an SNV-SVM model to be optimal for rice classification, also reaching 100% accuracy. Feng et al. [11] also benchmarked several models for maize leaf screening and found that algorithms like Extreme Learning Machine (ELM) and Radial Basis Function Neural Network (RBFNN) could achieve prediction accuracies over 90%.

2) Deep Learning Approaches: More recently, deep learning models, particularly Convolutional Neural Networks (CNNs), have been explored for spectral analysis. These models offer the advantage of automatic feature extraction, potentially overcoming the limitations of traditional methods that rely on manual "feature engineering" [12].

A study by Wei et al. [12] directly compared deep learning with traditional methods for GM maize kernel classification. By transforming the 1D spectra into 2D matrices, their modified VGG Net (a CNN architecture) achieved a prediction accuracy of 96.1%, significantly outperforming a backpropagation neural network (BPNN) model built on PCA-extracted features (73.3% accuracy). This indicates that deep learning models are highly effective at learning the intricate, synergistic relationships between different spectral bands. Similarly, Sohn

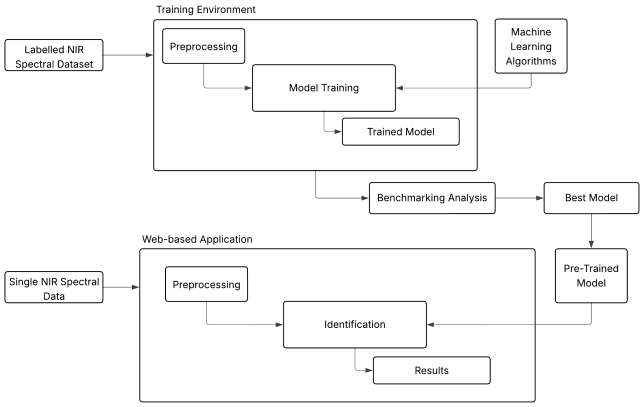

Fig. 1. The proposed data analysis pipeline for model development and identification.

et al. [13] reported a high classification accuracy of 98.9% using a CNN for discriminating canola varieties and their hybrids. These findings highlight the significant potential of deep learning to enhance the accuracy and robustness of NIR-based GMO detection systems.

III. Materials and Methods

The computational framework for this study will be designed to systematically develop, benchmark, and deploy a machine learning model for the non-destructive identification of Bt corn from non-Bt corn using NIR spectral data. The proposed framework, which is illustrated in Figure 1, will be divided into two primary phases: (1) an offline process for model development and benchmarking, and (2) a conceptual workflow for an online identification system.

A. Data Acquisition and Preparation

1) Sample Collection and NIR Spectral Acquisition: The dataset for this study will consist of NIR spectra from both transgenic and non-transgenic (non-Bt) corn kernels (*Zea Mays*). The samples will be sourced from the Institute of Plant Breeding (IPB) at the University of the Philippines Los Baños (UPLB). A total of [NUMBER] kernel samples will be obtained, comprising [NUMBER] Bt corn samples and [NUMBER] non-Bt corn samples.

The NIR diffuse reflectance spectra will be acquired using a Thermo Antaris II FT-NIR Analyzer (Thermo Fisher Scientific, Waltham, MA, USA). Spectra will be collected in reflectance mode over the full wavelength range of the instrument, typically 4000cm^{-1} to 10,000 cm^{-1}. For each kernel, spectra will be recorded at a resolution of [], and [] scans will be co-added and average to produce a single, high signal-to-noise ratio spectrum. This process will yield a Labelled NIR Spectral Dataset, where each spectrum is tagged with its corresponding class (Bt or non-Bt).

The raw spectral data will be exported in the .spa file format. All subsequent data handling and analysis will be conducted in a Python environment, where the SpectroChemPy library will be used to import the .spa files and convert the spectral data into a numerical format.

2) Spectral Preprocessing: Raw NIR spectra are often affected by physical phenomena such as light scattering, baseline shifts, and instrumental noise, which can obscure the underlying chemical information [14]. To reduce these effects and create a consistent input for all models, a single, standardized preprocessing technique will be applied.

This study will employ Savitsky-Golay (SG) smoothing. This method was chosen for its proven effectiveness in reducing high-frequency noise from spectral data while preserving the essential shape and width of the absorption peaks [14]. A second-order polynomial with a window size of 11 data points will be applied to all spectra. By using a fixed preprocessing step, this study will ensure that any observed differences in performance are directly attributable to the machine learning algorithms being benchmarked.

3) Dataset Partitioning: The preprocessed dataset will be partitioned into a training set and a held-out test set. A stratified random split will be employed to maintain the original proportion of Bt and non-Bt samples in both subsets. An 80:20 ratio will be used which will result in approximately [NUMBER] samples for training and [NUMBER] samples for testing. The training set will be used exclusively for model development and hyperparameter tuning, while the test set will be reserved for the final performance evaluation.

B. Offline Model Development and Benchmarking

This phase will focus on training and comparing different machine learning models to identify the most effective classifier for this task. All development will be conducted in Python [VERSION] using the `scikit-learn` [VERSION] and `TensorFlow` [VERSION] libraries.

1) Machine Learning Algorithms: A selection of Machine Learning Algorithms, representing both traditional and deep learning approaches, will be benchmarked:

1) **Partial Least Squares Discriminant Analysis (PLS-DA):** As a classical supervised chemometric method, PLS-DA is well-suited for high-dimensional and collinear spectral data. The optimal number of latent variables (LVs) will be determined using 10-fold cross validation on the training set.
2) **Support Vector Machine (SVM):** An SVM with a Radial Basis Function (RBF) kernel will be implemented because it is highly effective for complex, non-linear classification problems. The hyperparameters ‘C‘ (regularization parameter) and the kernel coefficient ‘gamma‘ will be optimized using a grid search with 10-fold cross-validation.
3) **1D Convolutional Neural Network (1D-CNN):** A deep learning model will be designed to automatically learn hierarchical features from the 1D spectral data which may capture complex patterns missed by traditional methods. The proposed architecture will consist of:
   - Two 1D convolutional layers with [NUMBER] filters.
   - A max pooling layer following each convolutional layer.

- A flatten layer followed by two dense (fully-connected) layers.
- A final output layer with a sigmoid activation function for binary classification.

The model will be trained using the Adam optimizer with a binary cross-entropy loss function.

2) Benchmarking Analysis and Model Selection: The performance of each Trained Model will be evaluated on the held-out test set. A comprehensive Benchmarking Analysis will be conducted by comparing the models based on the performance metrics defined in Section III-E. The model that demonstrates the highest F1-score and overall robust performance will be selected as the Best Model. This final model will be saved as the Pre-Trained Model for the online identification system.

C. Online Identification Workflow

The second part of the framework will outline the practical application of the developed model in a real-world scenario. An end-user will be able to upload a Single NIR Spectral Data file from an unknown corn kernel. This spectrum will automatically undergo the standardized Preprocessing pipeline identified as optimal during model development. The preprocessed spectrum will then be fed into the Identification module, which will load the Pre-Trained Model to perform classification. The module will output the final Results: the predicted class (Bt or Non-Bt) and a corresponding confidence score.

D. Development of the Online Identification System

To translate the offline model into a practical tool, a prototype web-based application will be developed for the online identification workflow. This tool will provide an accessible and easy-to-use interface for end-users to classify unknown corn samples without requiring specialized programming knowledge.

1) Implementation Pipeline and Technologies: The application will be built using a client-server architecture, as illustrated in Figure 2. **The backend will be developed in Python using the FastAPI framework**. The frontend will be constructed with HTML, CSS, and JavaScript.

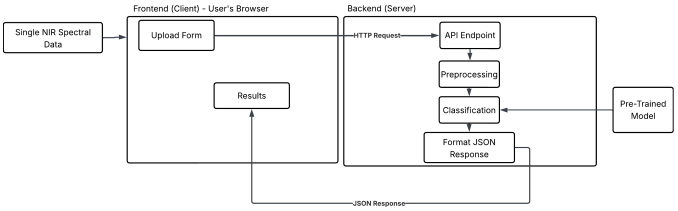

Fig. 2. Architectural diagram of the online identification system that shows the interaction between the frontend client and the backend API

The implementation pipeline will follow these steps:

1) **Model Export:** The final, best-performing model selected during the benchmarking phase (the Pre-Trained Model) will be saved to a file. For the scikit-learn models (PLS-DA, SVM), this will be a `.pkl` file using the `pickle` library. For the 1D-CNN, the model will be saved in the Keras `.h5` format. This file represents the direct output of the offline training process that will be loaded by the online system.
2) **Backend API Development:** A FastAPI server will be created with an API endpoint that accepts a file upload (the NIR spectrum of an unknown sample).
3) **Backend Processing:** Upon receiving the file, the backend will:
   - Load the Pre-Trained Model file.
   - Read and parse the uploaded spectral data file.
   - Apply the same standardized Savitzky-GOlay preprocessing used during training.
   - Feed the preprocessed spectrum into the loaded model to obtain a prediction.
   - Return the classification result and a confidence score in a JSON format.
4) **Frontend User Interface:** The frontend will feature a simple user interface allowing a user to upload a spectral data file. After submission, it will send the file to the backend API, wait for the response, and then clearly display the classification result to the user.

   2) Usability Testing: To ensure the tool's effectiveness and user-friendliness, a preliminary usability test will be considered. A small group of potential end-users, such as students or researchers in agricultural science, would be tasked with using the web application to classify a set of test spectra. Feedback would be gathered through a short survey or a think-aloud protocol to assess the system's ease of use, the clarity of the presented results, and overall user satisfaction. The insights from this testing would guide future improvements to the application's 9interface and workflow

E. Performance Evaluation Metrics

To quantitatively assess and compare the models, a confusion matrix will be generated for the test set predictions. From this matrix, the following standard metrics will be calculated:

- **Accuracy:** The proportion of total correct predictions.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

- **Precision:** The model's ability to correctly identify only the relevant instances.

$$\text{Precision} = \frac{TP}{TP + FP}$$

- **Recall (Sensitivity):** The model's ability to find all relevant instances.

$$\text{Recall} = \frac{TP}{TP + FN}$$

- **F1-Score:** The harmonic mean of Precision and Recall, providing a single score that balances both metrics.

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

where TP, TN, FP, and FN represent True Positives, True Negatives, False Positives, and False Negatives, respectively, with "Bt corn" defined as the positive class. A statistical

comparison, such as McNemar's test, will be used to determine if performance differences between the top-performing models are statistically significant (p ¡ 0.05).

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